Lesson 7: Records and Classes: Virtual Pet

Module 2 | 60 minutes | lists of dicts, classes | Grades 7–11 · ages 12–16

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. record ___ A. One real thing built from a class, like Pet("Moti").
b. list of dictionaries ___ B. A blueprint that describes data and actions together.
c. class ___ C. A table of records: one dictionary per row.
d. object ___ D. One item of data with several parts, like one player's dictionary.

2. Plan your program

Before you type it: Team manager functions. Write total_runs(team) that returns the total runs of every player in the list, and top_scorer(team) that returns the NAME of the player with the most runs. Each player is a dictionary with "name" and "runs" keys. Write the steps in plain English (pseudocode), then the Python you will use.

3. Predict the output: the batting object

What does this program print? Code: class Player: ⏎ def __init__(self, name): ⏎ self.name = name ⏎ self.runs = 0 ⏎ ⏎ def hit(self, r): ⏎ self.runs += r ⏎ ⏎ p = Player("Arjun") ⏎ p.hit(4) ⏎ p.hit(6) ⏎ print(p.name, p.runs)

a. A. Arjun 10
b. B. Arjun 6
c. C. Arjun 0
d. D. p 10

4. Build the blueprint

Put the lines of a class program in order Number them 1–6.

a. ___ moti = Pet("Moti")
b. ___ def feed(self):
c. ___ self.name = name
d. ___ def __init__(self, name):
e. ___ class Pet:
f. ___ moti.feed()

5. Plan your program

Before you type it: Debug it! This program should print `Moti 3` (the pet's name and its hunger after one feed). It has two bugs about self. Fix them. Write the steps in plain English (pseudocode), then the Python you will use.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can store and loop over records in a list of dictionaries.
☐ I can find a total and a top value in a list of records.
☐ I can write a class with __init__, attributes and methods.
☐ I can create objects and call their methods.